ZIG 0.18.0-DEV COMPATIBILITY PATCH

TARGET: 0.18.0-DEV.146+35ACCC06E

# OXWM — Zig 0.18.0-dev Migration Guide

Complete summary of build system, C translation, language syntax, and standard library updates required to compile OXWM on modern Zig.

| COMPILER VERSION | NEW HEADER FILE | MODIFIED FILES | BUILD STATUS |
|---|---|---|---|
| `0.18.0-dev.146` | `src/c.h` | `9 Zig Files` | `Passing (zig-out)` |

## 1. C Header Migration (@cImport Removal)

`src/c.h + 3 files`

Zig removed the built-in `@cImport`, `@cInclude`, and `@cDefine` functions in favor of build-system-managed C translation. All C headers required across OXWM were consolidated into a single `src/c.h` file and imported via a module named `"c"`.

**New File: src/c.h** — Created

```
#ifndef OXWM_C_H
#define OXWM_C_H

#include <time.h>
#include <lua.h>
#include <lualib.h>
#include <lauxlib.h>
#include <X11/Xlib.h>
#include <X11/Xutil.h>
#include <X11/Xatom.h>
#include <X11/keysym.h>
#include <X11/cursorfont.h>
#include <X11/extensions/Xinerama.h>
#include <X11/Xft/Xft.h>
#include <fontconfig/fontconfig.h>

#endif
```

**Zig Source Imports** — 3 Files Updated

```
// 1. src/bar/blocks/datetime.zig (Line 3)
// Replaced @cImport({
@cInclude("time.h"); })
const c = @import("c");

// 2. src/config/lua.zig (Line 10)
// Replaced @cImport({ lua.h, lualib.h,
lauxlib.h })
const c = @import("c");

// 3. src/x11/xlib.zig (Line 1)
// Replaced pub const c = @cImport({ X11
headers... })
pub const c = @import("c");
```

## 2. Build Script Updates (b.addTranslateC)

`build.zig`

Updated `build.zig` to translate `src/c.h` using `b.addTranslateC`, include the Lua dependency headers and system `/usr/include/freetype2` path (required by `Xft.h` for `ft2build.h`), and wire the resulting `c_mod` into the main executable and unit test modules.

**build.zig — C Translation & Module Wiring** — Updated

```
    // Translate C headers from src/c.h into a Zig module
    const translate_c = b.addTranslateC(.{
        .root_source_file = b.path("src/c.h"),
        .target = target,
        .optimize = optimize,
        .link_libc = true,
    });
    translate_c.addIncludePath(lua_headers);
    translate_c.addSystemIncludePath(.{ .cwd_relative = "/usr/include/freetype2" });
    const c_mod = translate_c.createModule();
    c_mod.linkSystemLibrary("X11", .{});
    c_mod.linkSystemLibrary("Xinerama", .{});
```

## 3. Array Repetition Operator (**) to @splat()

3 Files / 8 Lines

The binary array repetition operator ( `**` ) was removed from the Zig language, causing the parser to misinterpret `**` as a multiplication operator followed by a pointer dereference ( `error: binary operator '*' has whitespace on one side, but not the other` ). All 8 occurrences were replaced with `@splat()` .

| File & Line | Old Syntax (Removed in 0.18.0-dev) | New Syntax (@splat) |
|---|---|---|
| `src/config/config.zig:46` | ~~`[_]KeyPress{.{}} ** 4`~~ | `@splat(.{})` |
| `src/config/config.zig:204` | ~~`.{null} ** 12`~~ | `@splat(null)` |
| `src/keyboard/chord.zig:16` | ~~`.{config.KeyPress{}} ** max_chord_len`~~ | `@splat(config.KeyPress{})` |
| `src/monitor.zig:15` | ~~`[_]i32{1} ** 13`~~ | `@splat(1)` |
| `src/monitor.zig:16` | ~~`[_]f32{0.55} ** 13`~~ | `@splat(0.55)` |
| `src/monitor.zig:17` | ~~`[_]u32{0} ** 13`~~ | `@splat(0)` |
| `src/monitor.zig:18` | ~~`[_][6]?*const Layout{.{ null, ... }} ** 13`~~ | `@splat(@splat(null))` |
| `src/monitor.zig:19` | ~~`[_]bool{true} ** 13`~~ | `@splat(true)` |
| `src/monitor.zig:52` | ~~`.{null} ** 6`~~ | `@splat(null)` |

## 4. Standard Library API Updates

4 Files / 5 Lines

Several standard library helpers in `std.meta` , `std.mem.Allocator` , and `std.fmt` were deprecated or renamed in Zig 0.18.0-dev. In addition, `std.lang.Type.Enum` replaced its `.fields` slice with `.field_names` and `.field_values` .

**Standard Library Replacements Across src/** Updated

```
// 1. src/wm/actions.zig (Lines 394 & 430) — Enum reflection via @typeInfo
// Old: std.meta.fields(config_mod.Layouts).len
const new_lt = (monitor.sel_lt + 1) % @as(u32,
@intCast(@typeInfo(config_mod.Layouts).@"enum".field_names.len));
const lt_index = index % @as(u32,
@intCast(@typeInfo(config_mod.Layouts).@"enum".field_names.len));

// 2. src/config/lua.zig (Line 42) — Null-terminated formatted buffer print
// Old: std.fmt.bufPrintZ(&setup_buf, "package.path = '{s}/?.lua;' .. package.path", .{dir})
const setup_code = std.fmt.bufPrintSentinel(&setup_buf, "package.path = '{s}/?.lua;' ..
package.path", .{dir}, 0) catch return false;

// 3. src/bar/bar.zig (Line 69) & src/overlay.zig (Line 38) — Null-terminated string
duplication
// Old: allocator.dupeZ(u8, config.font) / allocator.dupeZ(u8, font_name)
const font_name_z = allocator.dupeSentinel(u8, config.font, 0) catch { ... };
const font_name_z = allocator.dupeSentinel(u8, font_name, 0) catch { ... };
```

## 5. Build & Installation Commands

Quick Reference

**Terminal Commands (Run from Project Root ~/oxwm)** CLI

```
# Standard debug build (outputs binary to zig-out/bin/oxwm)
zig build
```